SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme : B.E / B.Tech

Course Code : ITA04

Course Name : Statistics with R Programming

Duration : 1: 30 Hr

Date : _____ Max Marks: 50 Marks

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply R programming fundamentals including data structures, vectors, matrices, arrays, and class types to perform mathematical operations and manage data effectively.
CO2	Implement looping constructs, control structures, and recursive functions in R to solve computational problems with efficient program logic.

Answer all the Questions

1. A college maintains student information including student name, age, department, and semester marks. Using R programming, explain how variables and data types are declared and used to store this information. Construct suitable atomic vectors for student names and marks, create a list to combine different types of information, and create a matrix to represent marks in multiple subjects. Write the R statements and explain the data structures selected and the output produced.

(CO1) (BL3)

2. A data analyst is working with the following subject marks recorded for five students: 78, 45, 92, 33, and 61. Develop an R program to store the marks in a vector, identify the values greater than or equal to 50 using logical vectors, and calculate the total, average, maximum, and minimum marks. Extend the program by creating a 2 × 2 matrix and a small array, and demonstrate suitable operations on these structures. Explain the results of the operations.

(CO2) (BL4)

3. A campus application must categorize student scores as "Excellent" (80 and above), "Pass" (50 to 79), or "Fail" (below 50). Write an R program using if-else statements to classify a given score. Then use a loop to process the scores 72, 38, 85, 49, and 91 and display the category for each score. Define a function that accepts a score and returns its category, and explain how the control structure and function work. (CO2) (BL3)

4. A department needs to maintain a student dataset containing Student_ID, Name, Department, and Mark. Design an appropriate normalized data frame in R using at least five sample student records. Write R statements to display the structure and summary of the data frame, select the Name and Mark columns, retrieve students scoring 70 or more, and add a Result column based on a pass mark of 50. Explain the purpose and output of each operation. (CO3) (BL3)

5. A library wants to maintain book records in a CSV file with the columns Book_ID, Title, Category, and Copies. Write an R program to create a data frame with at least five book records, save it to a CSV file, read the file back into R, display the records, filter books with more than two copies, and update the number of copies for a selected book. Demonstrate the relevant file input/output and data-frame operations, and explain how the updated data can be saved to a new CSV file.

- (10) (CO3) (BL3)
- Edges: {(A, B), (A, C), (B, C), (B, D), (C, D)}

Analyze the given campus route network using the backtracking approach. Construct the possible search paths, identify the choices that lead to successful or unsuccessful paths, and justify whether a route exists that visits every location exactly once. ***(CO2) (BL4)***